Master thesis

Thesis in mechanical engineering, sustainable design and innovation, 30 credits

3D-Printed and Metallised WR-12 Waveguide Prototypes for 77 GHz Radar Applications

Development of a Cost-Effective Manufacturing Process through Additive Manufacturing and Metallisation

Mechanical Engineering

Halmstad, 2026-06-18

Kosovare Fetahu

HÖGSKOLAN I HALMSTAD

Abstract

Polymer-based additive manufacturing combined with metallisation is a promising route for the rapid prototyping of waveguide components at millimetre-wave frequencies. This thesis develops and evaluates a cost-effective manufacturing process for WR-12 split-block waveguides at 77 GHz, intended to serve as a foundation for the future development of additively manufactured antenna prototypes. Seven waveguide prototypes, including straight sections, single 90-degree bends with three different radii, and S-bends with two different separation distances, were designed and simulated in Ansys HFSS, fabricated by stereolithography (SLA), coated with 5 μm of aluminium by physical vapour deposition (PVD), characterised by white-light interferometry following ISO 21920, and measured using a vector network analyser (VNA). The measured return loss exceeded 24 dB for every prototype, comfortably satisfying the 15 dB matching criterion, but the insertion loss ranged from 3.1 dB to 17.4 dB at 77 GHz, exceeding the 1 dB target. The measured surface roughness of the metallised channels was comparable to the aluminium skin depth at 77 GHz, and the residual resin observed in the longer, more complex channels is identified as the dominant cause of the spread in insertion loss between the prototypes. The findings show that polymer-based additive manufacturing is viable for compact 77 GHz waveguide components and provide a characterised baseline on which future antenna prototyping can build, while indicating that achieving lower insertion loss will require improved cleaning protocols and an alternative metallisation strategy such as electroplating.

Keywords: additive manufacturing; stereolithography; waveguide; automotive radar; 77 GHz radar; PVD metallisation; physical vapour deposition; WR-12 waveguide.

Preface

This thesis would not have been possible without the support and guidance of several people, to whom I would like to express my sincere gratitude.

First and foremost, I would like to thank my supervisor Ross Friel at Halmstad University for his invaluable guidance, constructive feedback, and encouragement throughout the course of this work. His expertise and dedication have been a constant source of motivation.

I would also like to extend my gratitude to my supervisor Haik Avetian at SafeRadar Research Sweden AB for his supervision and practical insights. A special thanks also goes to my colleagues Fredrik Kjörk and Ulf Carlberg, whose support, discussions, and willingness to help, made the day-to-day work considerably more enjoyable and rewarding. Working alongside the team at SafeRadar has been a truly enriching experience, and I am grateful for the opportunity to conduct my thesis in collaboration with them.

Finally, I would like to thank everyone at Halmstad University and SafeRadar Research Sweden AB who have in any way contributed to this project.

Table of Contents

1. Introduction....1
1.1 Background....1
1.2 Purpose and Objective....2
1.3 Limitations....3
2. Theory....4
2.1 Additive Manufacturing for RF and Waveguide Components....4
2.2 Polymer-based Additive Manufacturing Processes for Waveguide Fabrication....4
2.3 Metallization of Polymer Waveguide Components....5
2.4 Dimensional Tolerances at 77 GHz....5
2.5 State-of-the-art....6
3. Methodology....7
3.1 Research Approach....7
3.2 Choice of Methods....7
3.3 Assumptions and Simplifications....9
4. Implementation....10
4.1 Literature Review....10
4.2 Electromagnetic Design and Simulation....10
4.3 Additive Manufacturing and Post-Processing....13
4.4 RF Characterisation and Validation....17
5. Results....20
5.1 Electromagnetic Simulation Results....20
5.2 Fabrication Results....23
5.3 Surface Characterisation Results....24
5.4 Measurement Results....25
6. Discussion....27
6.1 Discussion of Results....27
6.2 Discussion of the Method....29
6.3 Future Work and Recommendations....30
6.4 Social, Economic, and Environmental Aspects....31
7. Conclusions....33
References....34
Appendices....37
Appendix 1....37
Appendix 2....38

1. Introduction

1.1 Background

The automotive industry is undergoing a comprehensive transformation where autonomous vehicles and advanced assistance systems (ADAS) are demanding progressively more advanced sensor technology. Several frequency bands have been investigated; however, there are two main frequency bands that are used in automotive radar systems, and they are the 24 GHz and 77 GHz bands. Radar sensors operating at 77 GHz have established themselves as a key component in these systems thanks to their ability to operate reliably under varying weather and lighting conditions (Waldschmidt et al., 2021; Hasch et al., 2012). Operation at 77 GHz also allows for an antenna of almost ⅓ of the size compared to a 24 GHz one. Another advantage of higher frequencies, such as 77 GHz, is their lower penetration into human skin, minimising tissue absorption and therefore making them a safer choice (Temga et al., 2024). With the development of next-generation radar systems, there is an increasing need for more advanced antennas with improved radio frequency (RF) performance, while cost efficiency, lightweight design and rapid product development are becoming increasingly critical factors.

Within this frequency range, WR-12 waveguides are used as standard components for transporting electromagnetic waves. The WR-12 waveguide has internal dimensions of 3.099 mm × 1.549 mm and covers the frequency range of 60-90 GHz (Waveguide Sizes | Dimensions & Cutoff Frequency - Everything RF, 2026). At 77 GHz, the wavelength corresponds to approximately 3.9 mm ($\lambda = c/f$), where λ is the free-space wavelength, c is the speed of light, and f is the frequency. At this high frequencies, manufacturing tolerances in the 50-100 μm range become critical for the electromagnetic performance of the components (Lomakin, 2022). This extreme precision is particularly challenging for WR-12 waveguides operating in the 76-81 GHz automotive radar band (Hasch et al., 2012; Temga et al., 2024). Traditionally, waveguide devices have been manufactured using high-accuracy subtractive machining, carving out the internal parts of the waveguide from a block of good electrical conductivity (typically aluminium or brass), thereby providing a very smooth surface and low power losses (Polo-López et al., 2021). These methods, such as the use of Computer Numerical Control (CNC) machining, are very expensive and result in highly manufactured antennas (Salzburg et al., 2023). The demand for antenna designs with challenging requirements has increased rapidly during recent decades, leading to a particular interest in additive manufacturing (AM). AM technology offers several advantages, including the ability to create complex geometries using diverse materials (Polo-López et al., 2021).

SafeRadar Research Sweden AB was founded in 2015 and is a product and research company based in Gothenburg, Sweden, specialising in advanced 4D scanning/imaging radar systems operating at 77/79 GHz (SafeRadar Research Sweden

AB, n.d). As a small company in a highly competitive industry, the ability to quickly iterate and test different antenna concepts is crucial for product development and innovation. External manufacturers currently charge a very high cost for new antenna designs, which severely limits the possibilities for iterative design and experimental development. This cost level makes it economically unreasonable to test different antenna configurations and optimize performance through physical prototypes. The possibility to quickly 3D-print and evaluate different antenna designs has therefore become a subject of interest.

Additive manufacturing, or 3D printing, has in recent years shown promising results for the fabrication of high-frequency RF components. AM offers several advantages compared to conventional manufacturing, including lower costs, shorter manufacturing times, reduced material usage, quick iteration between design versions and the freedom to create complex geometries that are quite difficult to manufacture through traditional methods. Stereolithography (SLA) has been proven particularly suitable for millimetre-wave applications, capable of achieving sufficiently high tolerances and enabling the fabrication of parts with excellent surface quality ($R_q \leq 1$ μm). SLA is, however, often limited to dielectric resins, necessitating additional metal plating strategies to achieve electrically conductive properties (Lomakin et al., 2019).

This thesis project is conducted as a collaboration between Halmstad University and SafeRadar Research Sweden AB. The project constitutes an opportunity to combine academic research with practical product development, where the goal is to establish a cost-effective internal manufacturing method for prototype antennas. By developing this capability internally, SafeRadar can significantly reduce both costs and lead times in their product development, enabling more innovative and experimental design solutions.

1.2 Purpose and Objective

The purpose of this thesis project was to develop and evaluate a cost-effective manufacturing method for waveguide structures at 77 GHz, based on additive manufacturing combined with metallization, as a foundation for SafeRadar's future in-house development of antenna prototypes. The work aimed to deliver a documented and verified manufacturing route that SafeRadar can adopt internally for prototype development.

Main objectives:

- Develop a complete manufacturing process for waveguide structures at 77 GHz with a target below 1,000 USD per prototype antenna

Sub-objectives:

- Conduct a literature review of additive manufacturing methods and metallisation processes for RF components.
- Design and simulate various test structures, ranging from simple straight waveguide sections to more complex bent geometries.

- Perform RF measurements (preferably with VNA equipment) and compare measured results against simulations in HFSS (High-Frequency Structure Simulator).

1.2.1 Problem definition

The central problem addressed in this thesis is that external manufacturing of prototype antennas at 77 GHz is both costly and slow, limiting SafeRadar's ability to iterate on antenna designs. In-house additive manufacturing combined with metallisation could offer a faster, cheaper and more sustainable alternative. However, future antenna prototypes will inevitably contain bends and routing structures, and the effect of bend geometry, such as the bend radius and the spacing between consecutive bends, on the RF performance of additively manufactured components at 77 GHz has not been systematically investigated. This thesis investigates waveguides with bends as a foundational step toward future in-house antenna prototyping and provides the technical basis on which SafeRadar can decide whether to adopt the process for waveguide structures that include such routing features.

Research questions:

1. Can polymer-based stereolithography, combined with vacuum aluminium metallization, produce split-block WR-12 waveguide components that achieve acceptable performance (return loss below -15 dB and insertion loss below 1 dB) at 77 GHz?
2. How does the bend geometry affect the RF performance of the waveguide?
3. To what extent does the manufacturing process, including the SLA print quality, surface roughness after metallization, and split-block assembly, influence the measured RF performance?

1.3 Limitations

To keep the project manageable within the framework of a 20-week thesis project, the following limitations have been made:

Frequency Range: The work focuses specifically on 77 GHz frequency, which is the primary frequency for SafeRadar´s applications.

Manufacturing methods: Primary focus on stereolithography (SLA); other polymer additive manufacturing methods, such as digital light processing (DLP) and fused deposition modelling (FDM), are not included in the work.

Metallisation method: Due to time limitations, an external company provided the metallisation of the prototypes.

Material Selection: Limitation to SLA resins available on the Formlabs website only.

2. Theory

2.1 Additive Manufacturing for RF and Waveguide Components

Additive manufacturing (AM), often referred to as 3D printing, builds objects layer-by-layer from digital models, in contrast to subtractive manufacturing methods that refer to the removal of material from solid blocks. According to Peverini et al.(2023), AM is leading to a major transformation across industrial sectors. In the space sector, e.g., as mentioned in the paper, it has evolved from producing structural parts such as fasteners to enabling the development of complete radio frequency (RF) components and payloads. This change has been made possible by steady improvements in the dimensional accuracy and surface smoothness that can be achieved with modern AM technology.

When it comes to AM for waveguide applications, the main advantage is the possibility to create lightweight components as well as complex internal structures, all in a single part. However, there are still concerns regarding the manufacturing accuracy and surface roughness that have proven critical (Lumia et al., 2022; Gold & Helmreich, 2017). This section reviews the principal polymer AM processes applicable to millimetre-wave waveguides, the metallization techniques required to achieve RF functionality, and the current state of the art for 77 GHz automotive applications.

2.2 Polymer-based Additive Manufacturing Processes for Waveguide Fabrication

2.2.1 Stereolithography (SLA)

The concept of stereolithography dates as early as the 1980s, and 1987 was the year when production with the first stereolithographic machines began, making them the first machines in the additive manufacturing field (Godec et al., 2022). Stereolithography is a so-called Vat Photopolymerization (VPP) process in which a UV laser selectively cures a liquid photopolymer resin layer by layer. After the parts are finished, there is a further solidification process that is done by UV curing in an oven. Epoxy resins are mostly used in the present, and they only exhibit low levels of shrinkage (1-2%), in contrast to acrylate resins, which were the first photopolymers developed, exhibiting a shrinkage of (5-20%). SLA also provides the highest dimensional accuracy among common polymer AM processes, typically ranging from 20 to 60 μm, with surface roughness (Ra) values around 1-3 μm (Lumia et al., 2023; Peverini et al., 2023)

2.2.2 Digital Light Processing (DLP)

The digital light processing is similar to the SLA process. However, it uses a digital projector screen composed of microscopic mirrors to selectively cure entire layers simultaneously, rather than tracing with a point laser (Peverini et al., 2023). DLP

offers comparable accuracy to SLA (20-60 μm) with potentially faster build times for small parts with fine features.

2.2.3 Fused Deposition Modelling (FDM)

Fused Deposition Modelling, also known as Fused Filament Fabrication (FFF), is a material extrusion process in which a thermoplastic filament is heated and deposited through a nozzle layer by layer (Lumia et al., 2022). FDM offers the widest material selection among AM processes; however, the process achieves significantly lower accuracy (100-200 μm) and rougher surfaces compared to the previous processes. It therefore limits its applicability at millimetre-wave frequencies where the waveguide dimensions are small and surface roughness becomes a significant fraction of the skin depth (Lumia et al., 2022). Nonetheless, FDM remains relevant for rapid prototyping at lower frequencies and for structural elements that do not carry RF signals.

2.3 Metallization of Polymer Waveguide Components

2.3.1 Overview of Metallization Approaches

Since 3D-printed polymer structures are non-conductive, a metallisation step is required to enable their use as RF waveguide components. Several methods have been investigated in the literature:

Electroless plating: Electroless plating is a chemical deposition process that uses chemical reactions and does not require an external electrical current. It provides a uniform conductive layer of metal (often silver, nickel or copper), often as a first step to provide a "seed" layer for electroplating (Kim et al., 2025; Le Sage, 2016). Key advantages include low equipment cost, no vacuum requirement and compatibility with a wide range of substrate materials. However, the process is relatively slow and involves potentially toxic chemicals.

Electroplating: Electroplating is one of the oldest metal deposition methods. It requires an application of voltage that makes the metal ions from the dissolved anode migrate through the solution and reduce onto the part's surface. However, electroplating, unlike electroless plating, is limited to conductive substrates only, requiring prior processes such as electroless plating (Kim et al., 2025).

Physical Vapour Deposition (PVD): PVD is a vacuum-based technique in which a substrate is bombarded by ionised argon gas in a high-vacuum chamber. Evaporated metal is then deposited onto the surface, providing high adhesion and uniformity (Kim et al., 2025). The principal drawbacks are the high equipment and operating costs due to the vacuum requirement.

2.4 Dimensional Tolerances at 77 GHz

Operation at 77 GHz places stringent requirements on the mechanical precision of a waveguide component. The free-space wavelength at this frequency is approximately 3.9mm. The WR-12 standard specifies internal cross-sectional

dimensions of 3.099 mm x 1.549 mm, with a recommended single-mode operating range of 60-90 GHz (Pozar, 2012).

For the waveguide walls, the electrical properties are dependent on the skin depth. For a good conductor at microwave frequencies, the metallisation thickness must exceed several skin depths. Skin depth δs is the distance at which the amplitude of an electromagnetic wave inside a conductor falls to 1/e (≈ 36.8 %) and is given by $\delta s = 1/\sqrt{(\pi f \mu_0 \sigma)}$, where $\mu_0 = 4\pi \times 10^{-7}$ H/m is the permeability of free space, f is the frequency, and σ is the electrical conductivity of the material (Pozar, 2012)

For aluminium ($\sigma \approx 3.5 \times 10\text{^}7$ S/m), the skin depth is approximately 0.31μm. For the wave to propagate without leakage through the conductor, the metallisation layer should therefore exceed 3-5 skin depths in thickness, corresponding to approximately 1-2 μm of aluminium layer. The skin depth also defines the characteristic length scale over which surface roughness influences conductor loss. Morgan (1949) showed that when the root-mean-square deviation of the surface is comparable to the skin depth, the power dissipation increases by approximately 60 % relative to that of a smooth conductor of the same conductivity.

2.5 State-of-the-art

Monolithic and split-block configurations of 3D printed waveguides in E- and D-band were presented by Quint et al. (2025). For one of the D-band split-block configurations metallised by electroless copper plating, an insertion loss between 2 and 3 dB was obtained from two different samples, as well as severe reflection fluctuations throughout the band. On the other hand, two similar D-band samples metallised by physical vapour deposition (PVD) followed by copper electroplating were tested, showing much better results, with an insertion loss between 0 and 1 dB and a return loss below -15 dB throughout the band. Furthermore, two E-band monolithic and slot-free waveguides showed promising results with an insertion loss between 0 and 0.5 dB and a return loss below -15 dB throughout the band. Shen et. al. (2017) presented waveguide prototypes in the W- and D-band, fabricated using stereolithography and an electroless metallisation technique, reporting a mean insertion loss of 0.16 dB/in (or 6.3 dB/m) for the measured W-band prototype. Ghazali et al. (2017) presented an X-band rectangular waveguide using SLA printing and copper electroplating. The waveguide was a split-block configuration with a surface roughness of around 3μm, showing promising results. D'Auria et al. (2015) presented a WR-10 split-block waveguide that was SLA printed and metallised by electroless plating, followed by copper electroplating. The waveguide showed a dissipative attenuation of 11 dB/m that was comparable to that of a commercial aluminium waveguide (10 dB/m). The literature shows that low-loss 3D-printed waveguide components are possible, but performance depends strongly on the metallisation process, resin, geometry, and assembly method. Few studies report a complete low-cost internal workflow for SLA-printed and metallised WR-12 prototypes evaluated for 77 GHz radar development. This thesis addresses that gap.

3. Methodology

3.1 Research Approach

This chapter describes the methodology employed to design, fabricate, and characterise a polymer-based additively manufactured waveguide antenna prototype operating at 77 GHz for automotive radar applications. The study follows a quantitative research approach, which Creswell & Creswell (2018) define as a strategy in which the researcher tests objective theories by examining relationships among measurable variables. In the context of this thesis, the measurable variables are the RF performance parameters of the fabricated waveguide, namely insertion loss, return loss, and bandwidth. These are then compared against simulated reference values to evaluate the viability of the chosen manufacturing process.

The methodology is structured in four sequential phases: (1) a literature review that establishes the theoretical and technological foundation of the study, (2) electromagnetic design and simulation, (3) additive manufacturing and post-processing, and (4) RF characterisation and validation. This phased approach was chosen due to each phase informing the next, making it ideal for this study.

3.2 Choice of Methods

3.2.1 Literature Review

Although the primary contribution of this thesis is the experimental design, fabrication, and measurement of a polymer-based waveguide prototype, the work is informed by a literature review that serves two purposes. First, it establishes the theoretical background necessary to make informed decisions about AM process selection, metallization techniques, and waveguide design at 77 GHZ. Second, it identifies the current state of the art and the performance benchmarks, with which the experimental results of this thesis are later compared.

The systematic literature review follows the approach described by Booth, Sutton and Papaiannou (2016) and is used to establish the current state of the art in additively manufactured waveguide components at millimetre-wave frequencies. The review identifies suitable AM processes, metallization techniques, and design strategies and provides the performance benchmarks against which the experimental results are compared. The search was carried out across IEEE Xplore, Scopus, and Google Scholar using combinations of keywords organised into three groups: manufacturing process terms, waveguide technology terms and application domain terms. While the primary interest is the 76-81 GHz band, the limited number of studies at this specific frequency necessitated broadening the scope to include work at other millimetre-wave frequencies.

3.2.2 Electromagnetic Design and Simulation

The first experimental phase involves electromagnetic (EM) design of the waveguide structure and prediction of its RF performance using full-wave simulation

software. The purpose of this phase is to establish the waveguide geometry that meets the requirements for the 76-81 GHz frequency, as well as to generate reference scattering parameters (S-parameter data) against which the measured performance of the fabricated prototype can be compared.

The full-wave electromagnetic simulations are used to design the waveguide structures, as well as generate reference S-parameter data. An alternative approach would have been to fabricate the prototypes directly and iterate based on measurement alone, but the simulation allows the design space to be explored efficiently through parametric sweeps before committing to fabrication, which is both time- and cost-effective. The simulation tool selected was Ansys HFSS 2024, which is an industry-standard finite element method (FEM) solver for high-frequency electromagnetic problems and is widely used in the published literature on waveguide components at millimetre-wave frequencies.

3.2.3 Additive Manufacturing and Post-processing

Stereolithography was selected as the polymer additive manufacturing process based on the findings of the literature review, as well as the available alternatives at Halmstad University (SLA and FDM). Among the polymer AM technologies evaluated, SLA offers the best combination of dimensional accuracy and surface roughness. FDM was rejected due to insufficient dimensional accuracy and high surface roughness, both of which would introduce unacceptable insertion loss at 77 GHz.

White-light interferometry was used to characterise the surface roughness of the metallised waveguide channels. This method was selected because it provides non-contact, three-dimensional surface profile measurements with sub-nanometre vertical resolution, which is necessary to resolve the surface features that influence conductor loss at 77 GHz. The alternative available method that was considered was contact profilometry; however, it was rejected because the stylus could damage the thin metallization layer and it is also impractical for the recessed internal surfaces of the waveguide channels. The surface measurements were performed after the metallization. Although measurements on the printed parts before metallization were performed, it was quickly noted that the transparent amber colour of the resin made the measurements unreliable. It was also deemed to be unnecessary due to the metallised surface being the one that actually interacts with the electromagnetic field.

3.2.4 RF Characterisation and Validation

Vector network analyser (VNA) measurement is the established method for characterising the S-parameters of passive microwave components and is used in this thesis to validate the fabricated prototypes. The measurement provides the empirical data against which the simulated performance is compared. A spectrum analyser was considered an alternative measurement instrument. However, a spectrum analyser only measures the magnitude of an incoming signal and does not provide reflection data, which means that the return loss (S11) of the waveguide cannot be determined. The VNA measures both reflection and transmission, captures both magnitude and phase,

and supports calibration techniques that remove systematic errors introduced by cables, connectors, and waveguide transitions. These capabilities are essential when measuring waveguides at 77 GHz (Pozar, 2012).

3.3 Assumptions and Simplifications

The electromagnetic simulations are performed as an idealised best-case scenario: the waveguide walls are modelled as a perfect electric conductor (PEC) with zero resistive loss, and the waveguide interior is modelled as a vacuum. No surface roughness is applied, and therefore, it is a deliberate simplification that establishes an upper bound on the achievable RF performance. Any degradation observed in the measurements can then be directly attributed to the manufacturing process, surface roughness, dimensional tolerances, finite conductivity and assembly imperfections, rather than to uncertainties and assumptions in the simulation models. Incorporating finite conductivity and surface roughness in the simulation models could provide closer agreement with the measurements, but it would require accurate knowledge of the metallization properties, which are not known beforehand and are part of what the thesis aims to characterise.

The waveguide design uses a split-block (E-plane split) configuration, in which the waveguide is divided into two halves along the broad wall. It is assumed that the two halves can be assembled with a negligible air gap at the split plane. In practice, any gap at the split introduces signal leakage and additional loss. This effect, however, is not modelled in the simulation but is expected to appear in the measurement results.

4. Implementation

This chapter describes the practical execution of each phase in chronological order. Technical details, including equipment, software settings, material properties, and process parameters, are documented to enable reproducibility (Bell & Waters, 2018).

4.1 Literature Review

The literature review was conducted following the systematic approach described by Booth, Sutton, and Papaioannou (2016), who emphasise the importance of a transparent and reproducible search strategy with clearly defined inclusion and exclusion criteria. The search was carried out across the electronic databases IEEE Xplore, Scopus, and Google Scholar, using combinations of keywords organised in three groups: manufacturing process terms (e.g., “3D printing”, “polymer”, “stereolithography”, “metallization”), waveguide technology terms (e.g., “waveguide”, “millimeter-wave”, “77 Ghz”, “antenna”), and the application domain (“automotive radar”). Boolean operators (AND, OR) were used to combine terms across groups, and the search was limited to English-language publications from 2010 onward, as polymer AM for millimetre-wave components is a relatively recent field. Studies were included if they reported quantitative RF performance data for waveguide components fabricated using at least one polymer-based AM process. While the primary interest of this thesis is the 76-81 GHz automotive radar band, the limited number of studies reporting results specifically at 77 GHz necessitated broadening the scope to include work conducted at other millimetre-wave frequencies. Priority was given to peer–reviewed journal articles and conference papers reporting quantitative RF performance data. The findings from the literature review are presented in the theory chapter and are referenced throughout the experimental phases to justify design choices and interpret measurement results.

4.2 Electromagnetic Design and Simulation

4.2.1 Waveguide Design

The waveguide design is based on the WR-12 standard rectangular waveguide, which has internal cross-sectional dimensions of 3.099 mm x 1.549 mm (width x height) and a recommended operating range of 60-90 GHz, covering the 76-81 GHz automotive radar band. To enable connection to standard E-band measurement equipment, each waveguide assembly is fitted with a UG-387/U precision flange at both ends. The UG-387/U is the standard flange for WR-12 waveguide and incorporates alignment pins that ensure precise mating with the corresponding flange on the VNA frequency extender, minimising the air gap and lateral misalignment at the interface, and thus minimising any error that could appear during measurement.

The experimental programme follows an incremental approach starting with simple straight waveguide sections, which serve as a baseline for evaluating the AM process and metallization quality in isolation from the geometric complexity. Once satisfactory performance is achieved with straight sections, the following prototypes will introduce

curved waveguide geometries that are more sensitive to manufacturing tolerances, beginning with a single 90-degree waveguide bend and progressing to a double-bend (S-bend) configuration. This staged approach allows the effects of geometric complexity to be separated from those of the fabrication process itself.

The straight waveguide section serves as the baseline configuration. It consists of a single rectangular waveguide of length L = 50mm with the standard WR-12 cross-section (see figure 4.1). This geometry contains no bends or transitions, and its simulated S-parameters therefore represent the best-case reference. Any deviation observed in the measured performance of a fabricated straight section can be attributed entirely to the manufacturing process, which includes the dimensional tolerances, surface roughness and metallization quality, without the geometric complexity as a confounding factor.

The single-bend geometry consists of two straight rectangular waveguide sections, with length L=50 mm, joined at a 90-degree angle. The bend is constructed by applying fillets to the inner and outer corners of the junction. The inner corner is filletet (rounded) with radius R, while the outer corner is filleted with a radius R + H, where H corresponds to the waveguide dimension in the plane of the bend: H = b for an E-plane bend, or H = a for an H-plane bend (see Figure 4.1). This parametric definition ensures that the waveguide cross-section is preserved throughout the bend, which is necessary to ensure correct wave propagation at 77 GHz. The bend radius R is treated as a design variable and is swept across a range of values to investigate its effect on the S-parameters, with the objective of identifying the minimum bend radius that yields acceptable return loss and insertion loss performance.

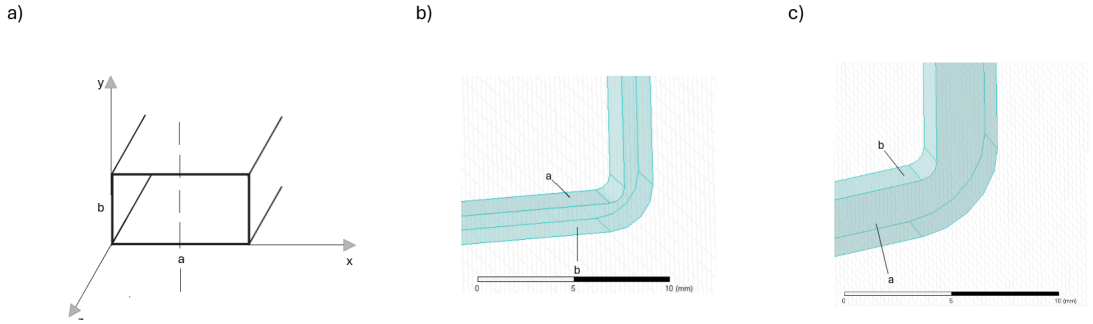


Figure 4.1: *Waveguide geometry definitions: (a) WR-12 rectangular cross-section showing the broad wall dimension a and the narrow wall dimension b ; (b) 90-degree bend in the E-plane, where the bend occurs in the plane containing the broad wall a; (c) 90-degree bend in the H-plane, where the bend occurs in the plane containing the narrow wall b.*

The double-bend geometry is an S-shaped waveguide with two 90-degree bends in opposite directions separated by a straight section of variable length D (see Figure 4.2). The separation distance between the two bends is introduced as a second design variable and is swept to study the coupling and interaction between adjacent bends.

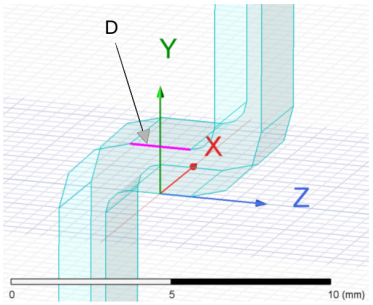


Figure 4.2: *Double-bended S-shaped waveguide design: Distance "D".*

4.2.2 Simulation Setup

All electromagnetic simulations are performed in Ansys HFSS 2024 using the modal network analysis solver, which computes the full-wave solution of Maxwell's equations in the frequency domain using the finite element method (FEM). The simulations are configured as an idealised best-case scenario, as described in Section 3.3.

The waveguide interior is assigned as vacuum, and the waveguide walls are modelled as perfect electric conductors (PEC). PEC assumes infinite conductivity, zero surface resistance, and a perfectly smooth surface. Two wave ports are defined on the rectangular cross-sectional faces at the input and output ends of the waveguide. The characteristic impedance at each port is set to Zpi, which is defined as the ratio of the voltage to the current weighted by the power flow. The Zpi definition is the standard choice for rectangular waveguide S-parameter normalisation. The excitation is a single-mode TE10 excitation at port 1, and the solver computes the reflected and transmitted waves at port 1 (S11) and port 2 (S21), respectively.

HFSS employs an adaptive mesh refinement process based on tetrahedral finite elements. The simulation domain is initially discretised into a coarse mesh, and the solver computes the S-parameters at the chosen solver frequency of 77 GHz. The mesh is then locally refined in regions where the field gradients are largest, in this case, the waveguide bends, and the S-parameters are recomputed. This adaptive cycle repeats until the convergence criterion is satisfied, meaning that the maximum change in any S-parameter magnitude between two consecutive passes (maximum delta S) must be less than or equal to 0.02 (the chosen value).

After the adaptive mesh had converged at 77 GHz, a frequency sweep between 60 and 90 GHz with 31 linearly spaced frequency points was performed. This covers the full WR-12 operating range and extends beyond the target band, allowing frequency shift caused by geometric deviations in the fabricated prototypes to be identified.

4.2.3 Parametric Studies

Two parametric studies were conducted. In the first study, the bend radius R of the single 90-degree bend is swept across the values 0, 0.4, 0.8, 1.0, 1.2, 1.6, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, and 30 mm to determine the influence on |S11| and |S21| across the

76-81 GHz band. The fine resolution at small radii (≤ 2 mm) is intended to capture rapid change in performance as the bend approaches a sharp 90-degree corner. The objective is to identify bend radii that maintain an acceptable return loss at the target band, since a tighter bend radius is desirable for compact antenna feed network layouts but is also more sensitive to manufacturing tolerances. The straight waveguide configuration is simulated at two lengths, L = 50 mm and L = 100 mm, to provide baseline reference data and to assess the effect of waveguide length on the measured transmission characteristics.

In the second parametric study, the separation distance D between the two bends of the S-bend configuration is swept across the values 0, 1, 2, 3 and 4 mm, with the bend radius fixed at R = 0.8 mm. This radius was selected based on the results of the first parametric study and is therefore treated as a fixed input rather than a free parameter in the second sweep. The separation sweep investigates whether closely spaced bends introduce additional resonances or coupling effects that degrade performance and determines the minimum interbend spacing required to avoid such effects. Both parametric studies generate families of S-parameter curves that serve as the reference data against which the measured performance of the fabricated prototypes is compared.

4.3 Additive Manufacturing and Post-Processing

The second experimental phase concerns the physical realisation of the waveguide design through polymer additive manufacturing and subsequent metallization. The fabrication process is documented in detail to enable reproducibility.

4.3.1 SLA Printing

The waveguide prototypes are fabricated using stereolithography (SLA) on a Formlabs Form 4B printer. The Form 4B is a masked stereolithography (MSLA) system based on Formlabs ´ Low Force Display (LFD) technology, in which a 4K liquid crystal display is illuminated by a backlit unit consisting of 60 uniform-wavelength LEDs at 405 nm with collimating lenses. The display selectively masks the UV light to cure each layer of photopolymer resin in a single exposure, regardless of the number of features on that layer. The printer offers an XY resolution of 50 μm (corresponding to the pixel size of the Light Processing Unit) and supports layer thickness ranging from 25 to 200 μm, depending on the material. The build volume is 200 x 125 x 210 mm (W x D x H). The Form 4B achieves a surface accuracy where 95% of the printed surface lies within 50 μm of the CAD model and 99.7% within 100 μm (Formlabs, n.d.).

The resin selected is Formlabs High Temp V2, an engineering photopolymer formulated for applications requiring high thermal stability. The material selection was driven by a critical constraint: the metallization step required the polymer substrate to withstand temperatures of at least 120 ℃ without dimensional deformation. Standard SLA resins have a heat deflection temperature (HDT) in the range of 50-75 ℃, which would risk warping or softening during metallization, potentially destroying the

dimensional accuracy achieved in the printing step. High Temp V2 has a heat deflection temperature of 238 ℃ at 0.45 MPa and 101 ℃ at 1.8 MPa, the highest HDT among all Formlabs resins. This provides a substantial thermal margin above the 120 ℃ metallization requirement. The material has a coefficient of thermal expansion (CTE) of 79.9 µm/m/℃ when fully cured, an ultimate tensile strength of 49 MPa, a tensile modulus of 280 MPa, and a flexural strength of 96 MPa. The resin has a transparent appearance and is compatible with the Form 4B at layer thicknesses of 50 and 100 µm (Formlabs, 2020).

A layer thickness of 50 µm was selected for all the waveguide prints. This is the finest layer thickness supported by High Temp V2 on the Form 4B and provides the best achievable surface finish in the build direction (Z-axis), which directly affects the internal surface roughness of the waveguide channels and therefore the conductor loss after metallization. The 50 µm setting also minimises the staircase effect on curved and angled surfaces, which is particularly important for the filleted bend geometries. The waveguide halves were oriented on the build platform with the surface opposite the internal channel selected as the base. This orientation ensures that no support structures are generated on either the internal channel surfaces or on the split-plane surface, where the two halves come in contact when assembled, both of which are critical for the RF performance. Support on the channel would directly affect the RF performance, and support on the split-plane surface would compromise the air-tight seal between the two halves when assembled. Support structures were generated automatically by PreForm with some adjustments to ensure that supports generated on the circular flange face of the UG-387/U interface were removed and relocated. This was done in order to maintain a flat and undamaged surface to ensure a proper interface with the VNA frequency extender. A representative print layout in PreForm is shown in Figure 4.3.

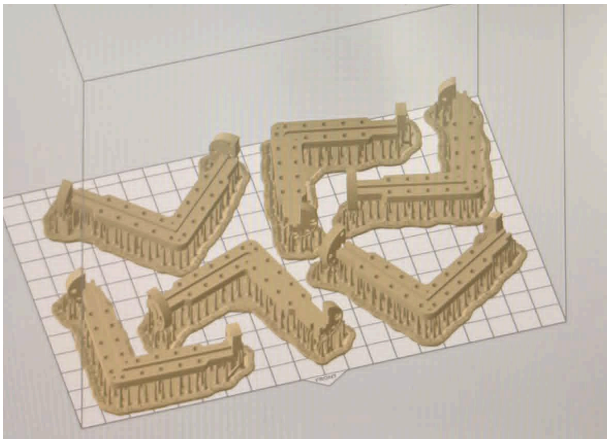


Figure 4.3: *Print layout in PreForm.*

4.3.2 Washing and Post-Curing

After printing, the parts were removed from the build platform and washed in isopropyl alcohol (IPA) for 5 minutes to remove uncured resin from the surfaces and internal channels. Formlabs specifies that High Temp V2 parts should not remain in

solvent for longer than six minutes, as excessive exposure degrades the cured material. After washing, the parts were air-dried and UV post-cured in a Formlabs Form Cure unit. To achieve the maximum HDT of 238 ℃, the manufacturers' recommended two-stage protocol was followed: first, UV curing for 120 minutes in the Formlabs Form Cure unit for 120 minutes at 80 ℃, followed by a thermal post-cure in a non-UV oven for 3 hours at 160 ℃. This two-stage process completes the cross-linking of the polymer network, which is necessary to realise the full thermal resistance. Given that the subsequent metallization step required temperatures of at least 120 ℃, full post-curing was essential to prevent dimensional deformation.

4.3.3 Metallisation

The metallization of the printed waveguide halves was carried out by AMB Industri AB (Broakulla, Sweden), a company specialising in vacuum metallisation of plastic components since 1973 (AMB Industri AB, n.d). Vacuum coating by physical vapour deposition (PVD) with evaporated aluminium was selected because it produces a uniform, fully metallic conducting layer on non-conductive polymer substrates.

The coating process is carried out in a high-vacuum chamber. In the first stage, the parts are bombarded with charged vapour ions to activate the polymer surface and create the conditions for strong adhesion of the subsequent metal layer. In the second stage, aluminium is heated to its evaporation temperature (approximately 1500 ℃) in a high vacuum and condenses onto the part surfaces.

A coating thickness of 5 μm of aluminium was specified for the waveguide prototypes in this study, corresponding to the upper end of the 3-5 μm process range used by AMB for EMC shielding applications (Hinz, 2018). The chosen thickness substantially exceeds the electromagnetic skin depth at 77 GHz (approximately 0.3 μm for aluminium), ensuring that the conducting layer is more than 15 skin depths thick. During the coating process, the part surface temperature reaches 115-120 ℃, which is the underlying reason that High Temp V2 was selected as the printing material rather than a standard SLA resin (see Section 4.3.1). The heat deflection temperature of 238 ℃ at 0.45 MPa for High Temp V2 provides sufficient thermal margin above this process temperature to ensure that the dimensional accuracy achieved during printing is not compromised by thermal deformation during metallisation.

4.3.4 Surface Characterisation

Surface topography and roughness of the metallised waveguide channels were measured using a Zygo NewView 9000 white-light interferometer. The measurements were performed after metallization for two reasons. First, the High Temp V2 resin is transparent and amber coloured in its cured state, which makes interferometric measurement of the uncoated polymer unreliable due to insufficient surface reflectivity and interference from sub-surface reflections. Second, and more importantly for this thesis, it is the metallised surface, and not the polymer substrate, that interacts with the electromagnetic field and determines the conductor loss at 77 GHz.

Measurements were performed on each fabricated prototype using a 5.5x objective with 1x zoom. To obtain an evaluation length sufficient for compliance with ISO 21920 for the specification of surface texture profile parameters, the stitching function in the Zygo Mx software was used in order to produce a single measurement of at least 5mm in length. For each prototype, stitched measurement locations were selected randomly within the metallised internal channel, since these surfaces are the ones that interact directly with the propagating electromagnetic field and therefore determine the conductor loss at 77 GHz.

The acquired surface data were subsequently processed in MountainsMap 2024, which is an industry-leading software for evaluating 3D measurement data. The processing workflow consisted of four steps. First, non-measured points were filled by interpolation from the surrounding measured points to produce a continuous surface dataset. Second, a region of 5 x 1 mm was extracted from each sample for analysis. Third, a multi-plane form removal operation using a third-degree polynomial was applied to remove waviness and long-wavelength form deviations from the surface. Fourth, a threshold operation based on height from the mean plane was applied to exclude any outliers or spikes from the analysis area.

From each sample, three parallel profile lines were extracted, and for each of the profiles, the arithmetic mean deviation of the assessed profile (Ra) and the root mean square (RMS or Rq in this case) were extracted in accordance with ISO 21920. The recorded roughness values are later correlated with the measured insertion loss to assess the extent to which surface roughness contributes to the total RF loss, relative to other loss mechanisms such as finite metal conductivity and split-block gap leakage.

4.3.5 Assembly

The waveguide prototypes were designed for operation at 77 GHz, corresponding to a free-space wavelength of $\lambda \approx 3.9$ mm ($\lambda/2 \approx 1.95$ mm). This wavelength served as the fundamental dimensional reference for all mechanical positioning decisions during assembly.

Each prototype was assembled by joining the individual waveguide sections and securing them with M2 stainless steel screws. Each hole was machined with a countersink sized to accommodate the M2 screw head flush with the surface. The screw hole pattern was defined with respect to $\lambda/2$. Holes were positioned 9 x ($\lambda/2$) ≈ 17.5 mm inward from each end flange, providing sufficient clearance from the flange interface while maintaining a mechanically stable anchor point. Along the length of the prototypes, successive holes were spaced 4 x ($\lambda/2$) ≈ 7.8 mm apart, ensuring uniform clamping pressure distributed at sub-wavelength intervals across the contact surface, as well as a sufficient amount of space between each mutter.

During assembly, screws were inserted from the front face and tightened progressively in a cross pattern to prevent differential compression or warping of the waveguide sections. The circular flanges at each end of the prototype, visible in the side profile, were used to align the assembly with the measurement setup (see Figure 4.4).

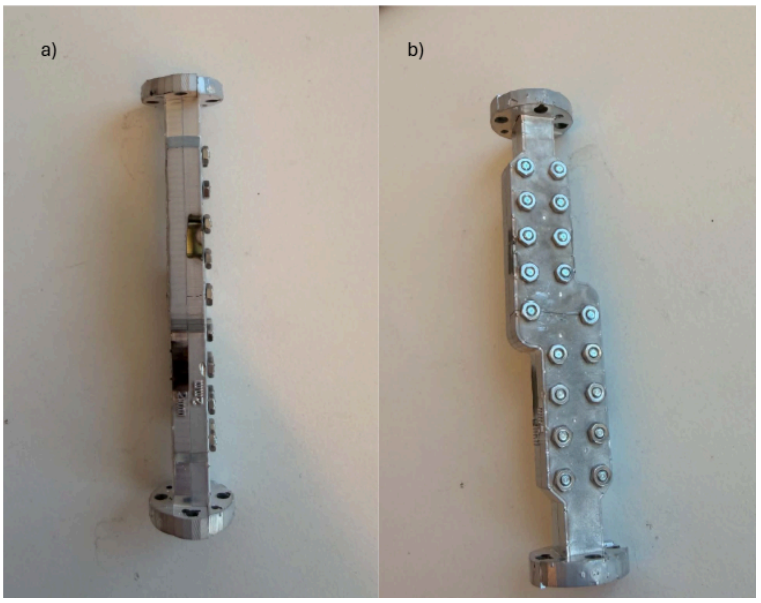


Figure 4.4: (*a) Side profile of an assembled prototype; (b) Back face of the same assembled prototype.*

4.4 RF Characterisation and Validation

The third and final experimental phase involves measuring the RF performance of the fabricated waveguide prototypes using a VNA and comparing the results with simulated reference data. This phase provides empirical evidence on which the conclusions of the thesis are based.

4.4.1 Measurement Equipment and Setup

RF characterisation of the waveguide prototypes was performed using a Rohde & Schwarz ZVA 24 VNA operating from 10 MHz to 24 GHz as the base instrument, extended into the W-band using two Rohde & Schwarz ZVA-Z110 frequency converter modules (75-110 GHz, WR-10 waveguide output). One converter module was connected to each port of the prototypes, with the WR-10 flanges of the converter mating directly to the prototype´s end flanges.

The frequency sweep was configured from 75 GHz to 110 GHz, covering the full W-band and encompassing the 77 GHz frequency of interest. Calibration was performed using the Rohde & Schwarz ZV-WR10 calibration kit (stock no.1307.7100.10), which is optimised for use with the ZVA-Z110 converter modules. The two ZVA-Z110 converter modules were placed on the measurement bench and aligned face-to-face with the prototypes inserted between them, as shown in Figure 4.5. Special screws belonging to the setup kit were used to secure the prototype flanges to the converter flanges.

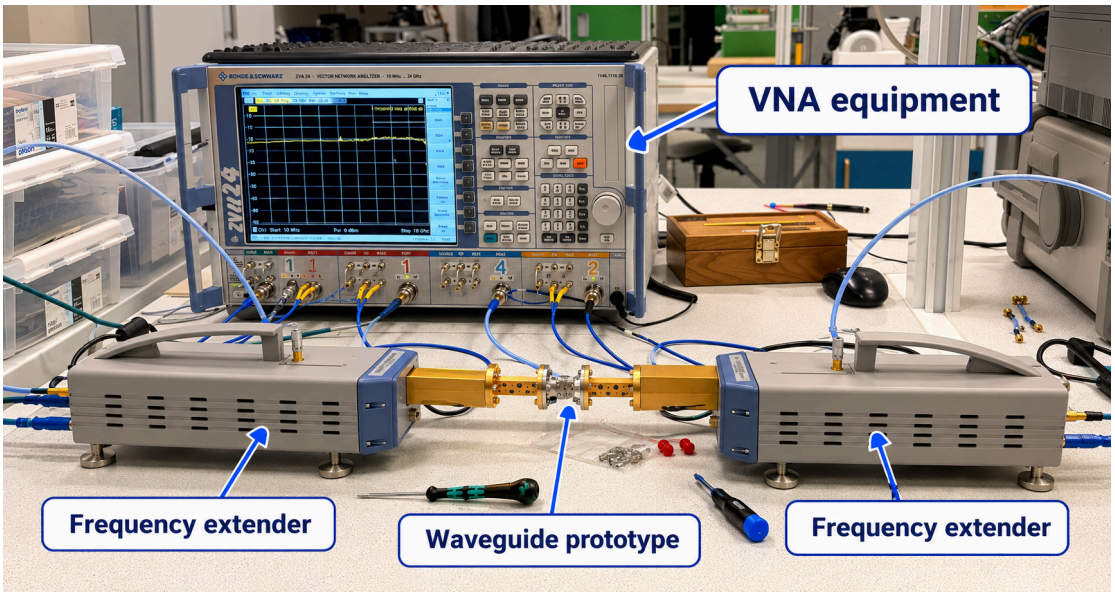


Figure 4.5: *Measurement setup.*

4.4.2 Calibration

A full two-port TOSM (Through-Open-Short-Match) system error correction was performed using the ZVA´s built-in Calibration Wizard, following the procedure described in the R&S ZV-WR10 technical information (document no. 1307.7174.92-4).

The calibration was carried out in the following sequence. For each converter port in turn, three one-port standards were measured: first, the Short standard, which also serves as the reflect standard. Secondly, the Offset Short, produced by inserting the calibration shim between the Short and the converter flanger, introducing a $\lambda/4$ phase shift at the centre of the operating band and a 180° phase shift of the reflection coefficient, and finally, the Fixed Match standard was used. This one-port sequence was repeated at both converter ports. A two-port "Through" measurement was then performed by connecting the two converter flanges directly to each other, without any intervening standard. Upon completion, the Calibration Wizard computed and applied the system error correction coefficients. The sliding match standard was not used in this work. While the sliding match can improve match accuracy, the fixed match was considered sufficient for the purpose of these measurements.

The reference planes after calibration were located at the WR-10 output flanges of the two ZVA-Z110 converter modules. All subsequent S-parameter measurements of the prototypes, therefore, include only the prototypes themselves, with the converter transitions, cables, and VNA systematic error removed by the calibration.

4.4.3 Data Acquisition and Analysis

The measured S-parameter data were recorded across the full W-band sweep from 75 GHz to 110 GHz, with the 77 GHz frequency being of primary interest. Following calibration, each prototype was mounted between the two ZVA-Z110

converter flanges and secured with the corresponding screws, ensuring contact at both interfaces (see Figure 4.5).

The S-parameter data were exported from the ZVA in a standard Touchstone format (.s2p) for post-processing. The following quantities were extracted for each prototype and will be presented in Chapter 5:

Insertion loss, defined as |S21| in dB, represents the total transmitted power loss through the prototypes relative to the calibrated reference.

Return loss, defined as |S11| in dB, represents the magnitude of the reflected signal at the input port.

5. Results

This chapter presents the outcomes of the work in four sections: the electromagnetic simulations used to design the waveguide geometries, the fabricated prototypes, the surface characterisation of the metallised waveguide channels, and the RF measurements of the assembled prototypes.

5.1 Electromagnetic Simulation Results

5.1.1 90-degree Bend Geometry

The bend radius R of the single 90-degree bend was swept across 17 values from 0 to 30mm. The full set of swept values is listed in Table 5.1.

Table 5.1. *Swept values of the bend radius R for the single 90-degree bend (in mm).*

R(mm)	0	0.4	0.8	1.0	1.2	1.6	2	3	4
R(mm)	5	6	7	8	9	10	20	30	—

Simulated |S11| and |S21| as a function of frequency over the 60-90 GHz range for selected bend radii are shown in Figure 5.1 and Figure 5.2 for the values 0, 0.4, 0.8, 1.0, 1.2 and 1.6 mm, and in Figure 5.3 and Figure 5.4 for the values 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, and 30 mm.

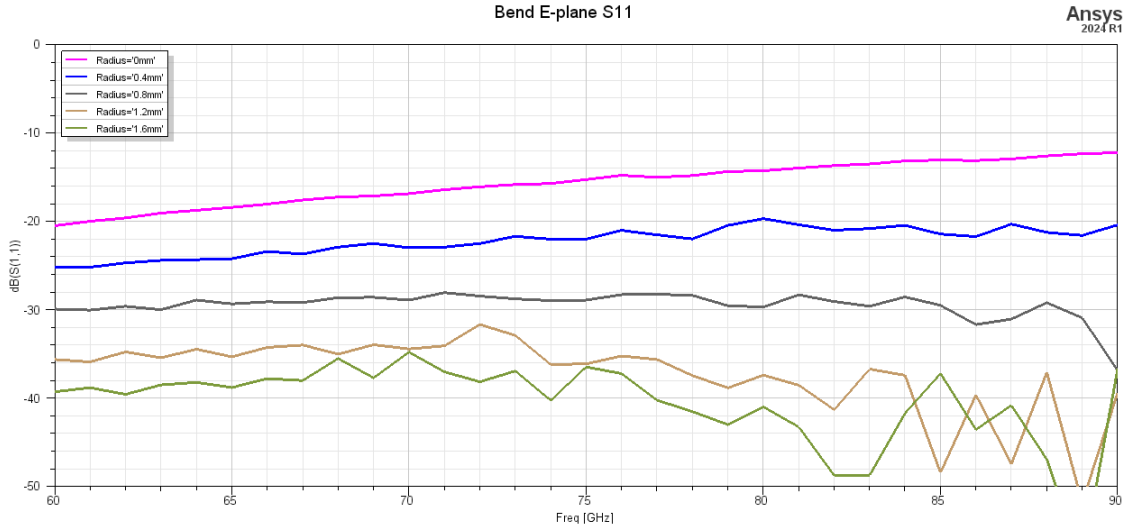


Figure 5.1: *Simulated |S11| for the R = (0, 0.4, 0.8, 1, 1.2 and 1.6 mm).*

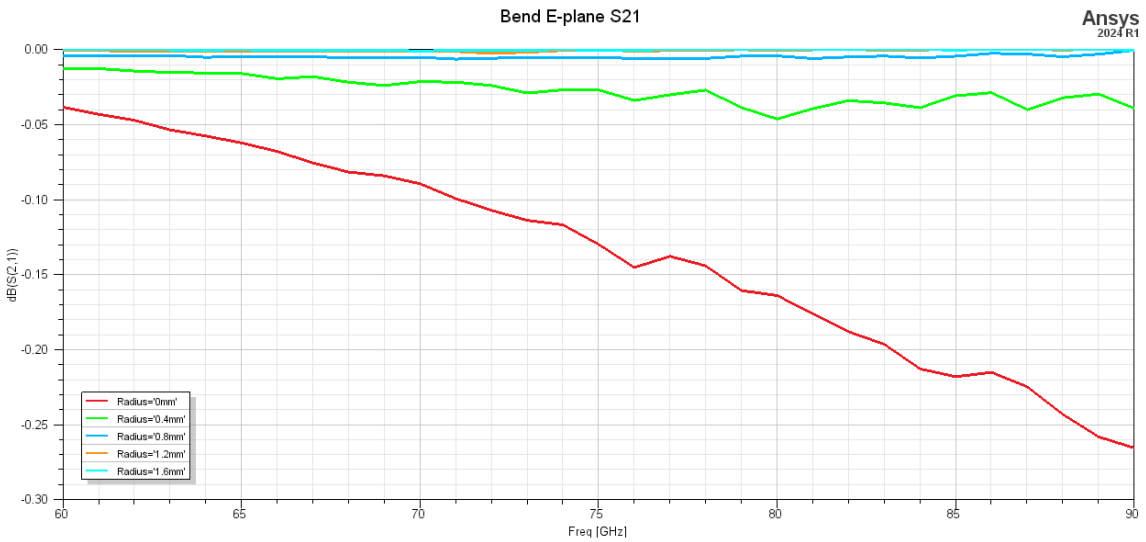


Figure 5.2: *Simulated |S21| for the R = (0, 0.4, 0.8, 1, 1.2 and 1.6 mm).*

For the simulated values in Figure 5.1 and 5.2, R = 0.8 mm showed the best simulated return loss and was therefore chosen for fabrication and as the fixed R-value for the following S-shaped configuration.

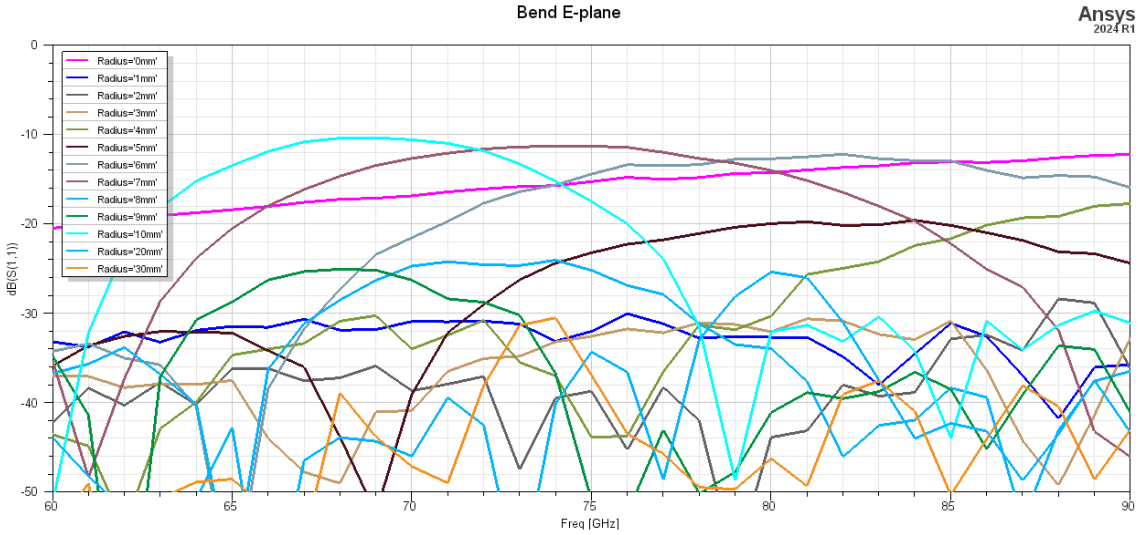


Figure 5.3: *Simulated |S11| for the R = (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, and 30 mm).*

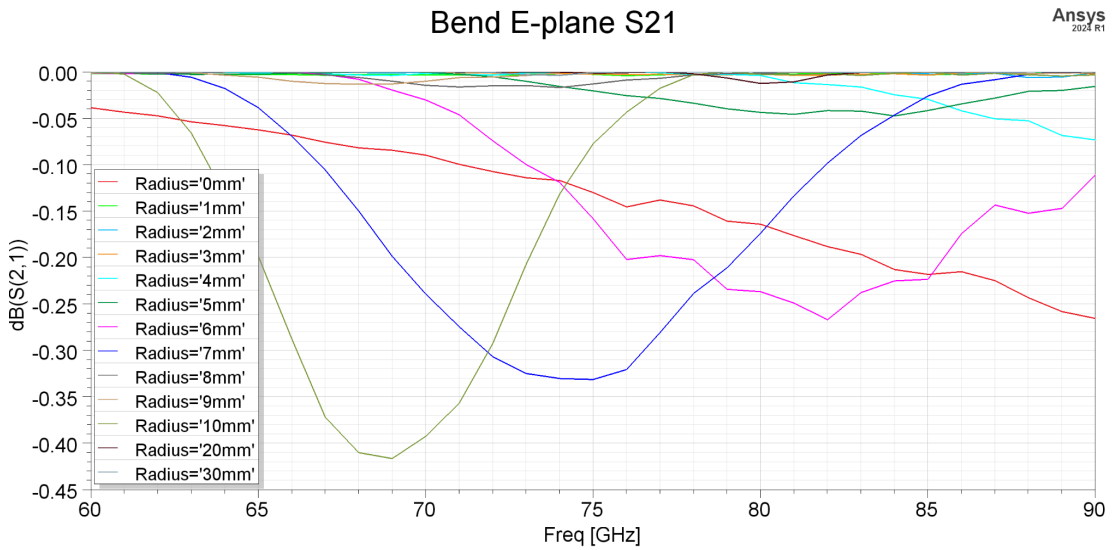


Figure 5.4: *Simulated |S21| for the R = (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, and 30 mm).*

The remaining chosen R-values (2 and 3 mm) were chosen based on their simulated return loss value of below 30 dB. R = 3 was chosen due to stability throughout the automotive frequency band (76-81) with minimal fluctuations, and R = 2 was chosen due to its return loss being exceptionally low (≈ 40 dB). As for the insertion loss, all simulated values were close to zero (best case scenario).

5.1.2 S-shaped Geometry

For the S-shaped geometry, the bend radius was fixed at R = 0.8 mm, and the separation distance D between the two bends of the S-bend configuration was swept across the values 0, 1, 2, 3, and 4 mm. The simulated S-parameters for each value of the distance D are shown in Figure 5.5 and Figure 5.6.

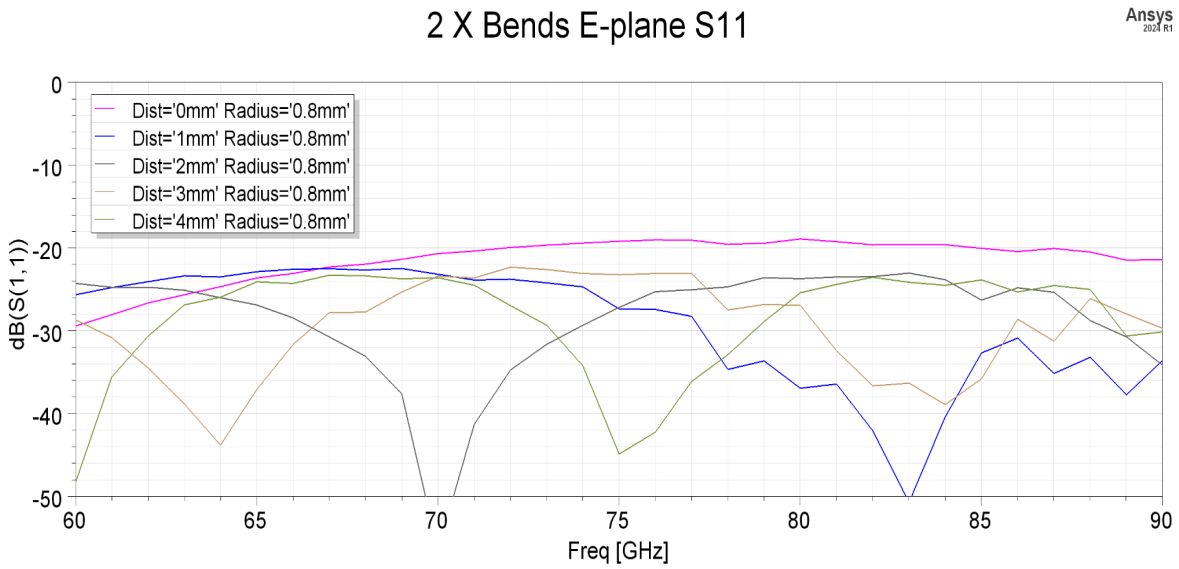


Figure 5.5: *Simulated |S11| for the distances D = (0, 1, 2, 3, and 4 mm).*

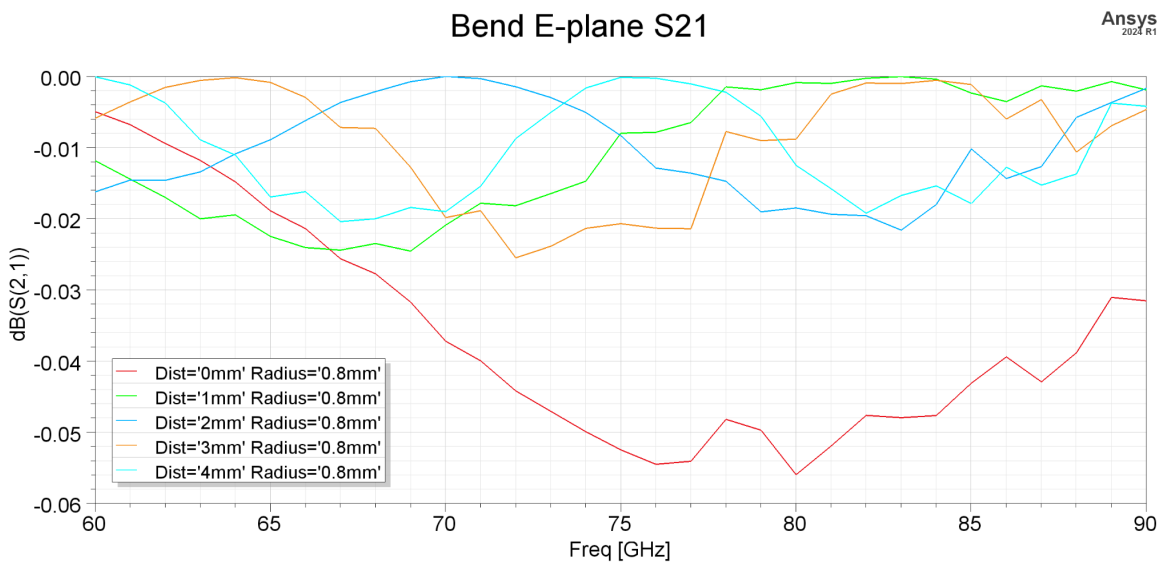


Figure 5.6: *Simulated |S21| for the distances D = (0, 1, 2, 3, and 4 mm).*

The chosen distances D were again based on the simulated values from the return loss, given that the insertion loss values all laid close to zero. The values D = 2 and 3 mm were chosen due to them exhibiting the most stable values (≈ 25 dB) in the automotive frequency band (76-81 GHz).

5.2 Fabrication Results

Seven waveguide configurations were fabricated based on the simulation results. The final configurations are summarised in Table 5.2.

Table 5.2. *Fabricated waveguide configurations.*

Configuration	Quantity	Length L (mm)	Bend Radius R (mm)	Separation Distance (mm)
Straight	1	50	—	—
Straight	1	100	—	—
Single 90° bend	1	50 + 50	0.8	—
Single 90° bend	1	50 + 50	2	—
Single 90° bend	1	50 + 50	3	—
S-bend	1	50 + 2 + 50	0.8	2
S-bend	1	50 + 3 + 50	0.8	3
Total number of prototypes	7	— — — — — — — — — — — — — — — — — —		

Photographs of the as-printed parts after post-curing and of the same parts after metallisation are shown in Figure 5.7 and Figure 5.8.

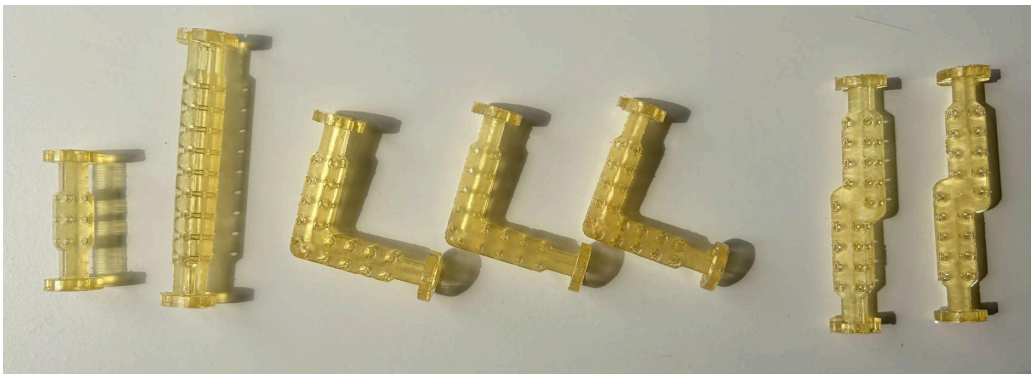


Figure 5.7: *Photograph of as-printed waveguides.*

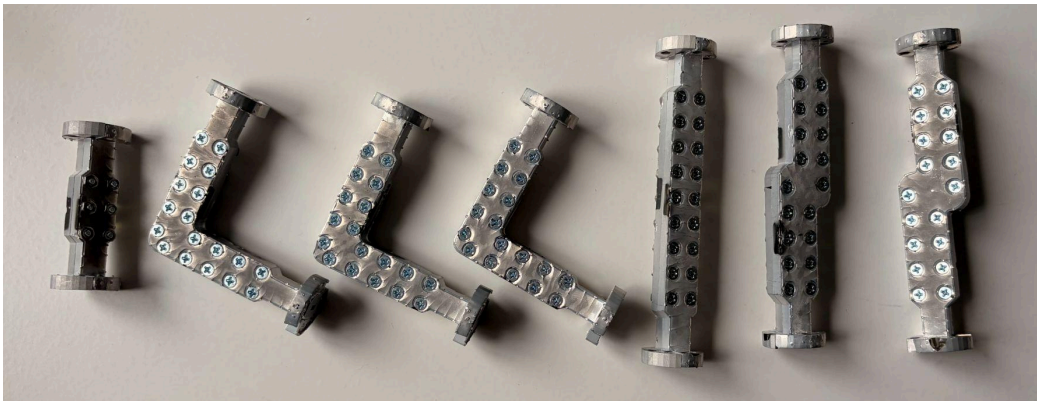


Figure 5.8: *Photograph of the metallised and assembled waveguides.*

5.3 Surface Characterisation Results

The Ra and Rq values extracted from three profiles per prototype are summarised in Table 5.3. Only the highest values (worst-case scenario) are reported in the table. Complete profile data are provided in Appendix 1.

Table 5.3. *Rq and Ra values of the metallised waveguide channels, calculated from one of the three profiles per prototype.*

Prototype	Rq (µm)	Ra (µm)
Straight - 50 mm	0.29	0.23
Straight - 100 mm	0.28	0.22
Single bend - R = 0.8 mm	0.29	0.22
Single bend - R = 2 mm	0.23	0.15
Single bend - R = 3 mm	0.24	0.17
S-shaped - D = 2 mm	0.29	0.24
S-shaped - D = 3	0.26	0.18

5.4 Measurement Results

The S-parameters of all seven fabricated prototypes were measured across the 75-110 GHz frequency range. The measured |S11| for all prototypes is shown in Figure 5.9, and the measured |S21| for all prototypes is shown in Figure 5.10. The measured parameters at the 77 GHz frequency are presented in Table 5.4.

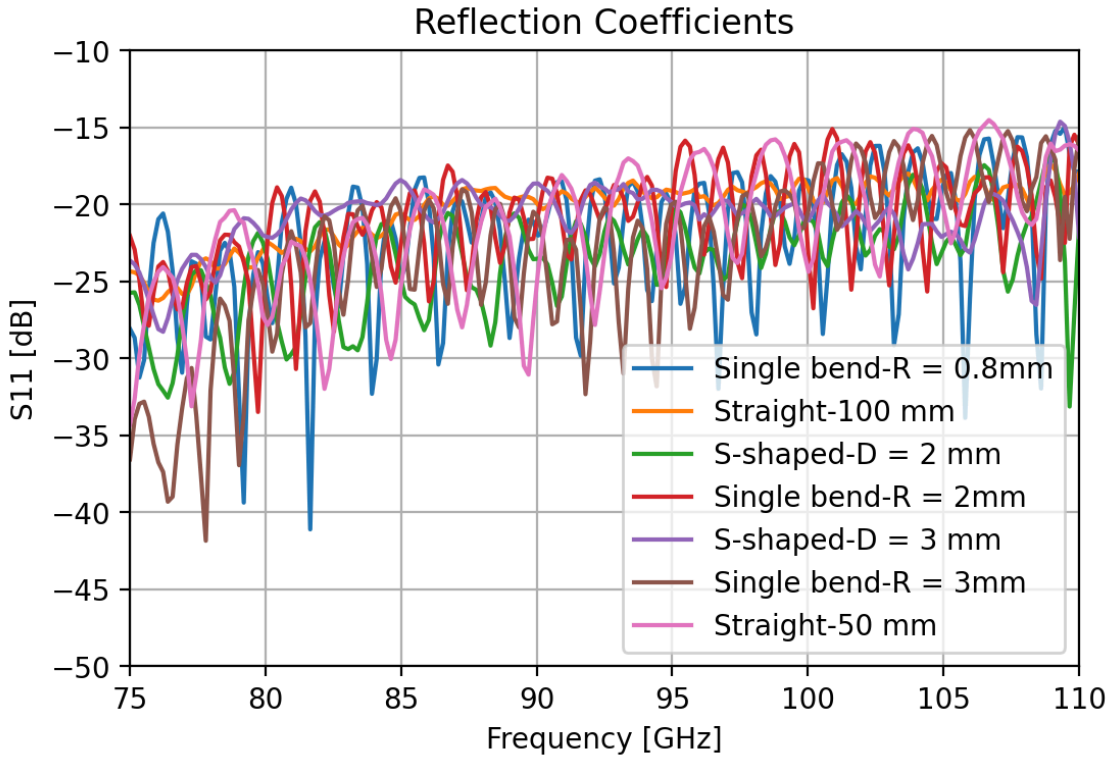


Figure 5.9: *Measured |S11| of all fabricated prototypes (75-110 GHz).*

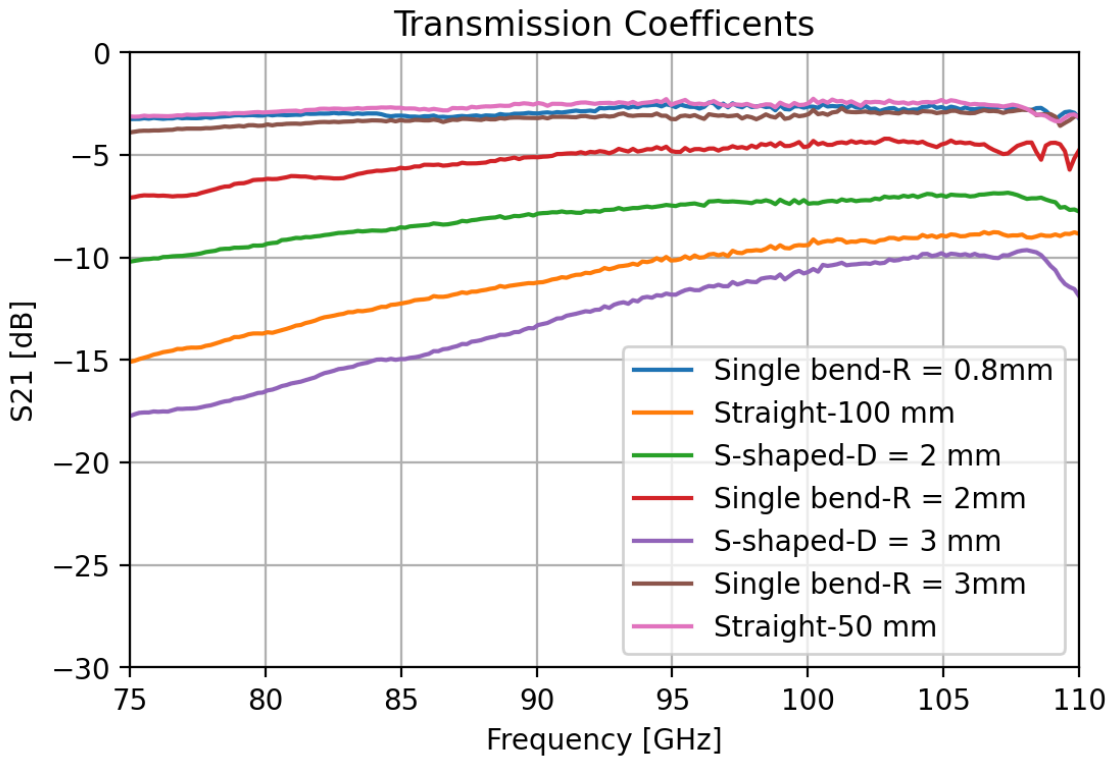


Figure 5.10: *Measured |S21| of all fabricated prototypes (75-110 GHz).*

Table 5.4. *Measured S-parameter values at the 77GHz frequency.*

S-parameter values at frequency 77 GHz		
	Return loss	Insertion loss
	S11	S21
Straight - 50 mm	-28.8 dB	-3.1 dB
Straight -100 mm	-25.8 dB	-14.4 dB
Single bend- R = 0.8 mm	-29.0 dB	-3.2 dB
Single bend-R = 2mm	-26.0 dB	-7.0 dB
Single bend-R = 3mm	-32.4 dB	-3.7 dB
S-shaped-D = 2 mm	-26.8 dB	-9.9 dB
S-shaped-D = 3 mm	-24.0 dB	-17.4 dB

6. Discussion

The aim of this thesis was to establish a technical foundation for SafeRadar's future in-house development of 77 GHz antenna prototypes by evaluating polymer-based stereolithography combined with vacuum aluminium metallisation as a manufacturing route. Rectangular waveguides containing bends were selected as the first "building block" because they isolate the effect of routing geometry, such as bend radius and inter-bend spacing, from the additional complexity of full antenna structures. This work addressed this aim in two parts: a parametric simulation study of bend geometries, for which no systematic data have previously been reported in the literature, and an experimental characterisation of seven fabricated prototypes spanning three configurations (straight, single 90-degree bend, and S-shaped).

6.1 Discussion of Results

The parametric simulation identified a bend radius of R = 0.8 mm as particularly favourable, yielding a simulated return loss of approximately -30 dB at 77 GHz. The S-bend separation was therefore swept with the radius fixed at R = 0.8 mm, showing that the distance between the bends only modestly influenced the simulated performance within the investigated range.

The measured return loss of all seven fabricated prototypes lay between -24 and -32.4 dB at 77 GHz, comfortably exceeding the -15 dB criterion set. The simulated return loss for the R = 0.8 mm bend (≈ - 30 dB) is closely matched by the corresponding measured value (-29 dB), and similar agreement is observed for the other configurations. From the perspective of the flange intersection matching at the waveguide ports, the prototypes meet the design requirements with a substantial margin, and the agreement between simulation and measurement of |S11| indicated that the geometric features responsible for matching are reproduced sufficiently accurately by the fabrication process.

The measured insertion loss, however, tells a different story. The seven prototypes span a wide range from -3.1 to -17.4 dB, and the dependence on geometry does not follow a simple monotonic pattern. The short straight section (50 mm) and the two single 90-degree bends (R = 0.8 mm and R = 3 mm) show the lowest insertion losses of -3.1, -3.2 and -3.7 dB, respectively, while the longer and S-bend configurations exhibit substantially higher losses. Notably, the 100 mm straight waveguide shows 14.4 dB of loss, more than four times the -3.1 dB measured loss for the 50 mm section. If the loss were dominated by the smooth-conductor ohmic loss in the metallised walls, the longer waveguide should exhibit approximately twice the loss of the shorter one. Therefore, the observed excess loss indicated that additional loss mechanisms accumulate along the length of the structure.

Several mechanisms can contribute to this behaviour. The Ra and Rq values of the metallised channels measured with the Zygo NewView 9000 yielded values comparable to the skin depth of aluminium at 77 GHz ($\delta \approx 0.31 \mu m$). Morgan (1949)

showed that the conductor loss is increased by approximately 60 % relative to a smooth surface when the root-mean-square (Rq) roughness is comparable to the skin depth, and therefore, the surface-roughness-induced loss is non-negligible. The split-block construction introduces a further loss mechanism where any physical gap at the split plane can act as a slot in the conducting wall and radiate as well as reflect (Pozar, 2012). The flange interface between the prototype and the VNA frequency extender may also be a possible source of additional transmission loss.

An additional process-related observation was made. During post-curing inspection of the printed waveguide halves, residual uncured resin was found inside the internal channels of the 100 mm straight, single bend R = 2 mm and the two S-bend prototypes (see Appendix 2). The resin residue, not being washed properly, is most likely due to contamination of the IPA bath after repeated washing. Resin residue inside the channel becomes embedded beneath the subsequently deposited aluminium layer, distorting the inner wall profile and degrading the smoothness and continuity of the conducting surface. The affected prototypes are precisely those that exhibited the highest measured insertion loss, and therefore, incomplete cleaning is seen to be a major contributor to the elevated insertion loss in these prototypes.

It is important to note that all seven prototypes were produced using identical systematic process parameters: the same Formlabs Form 4B printer, the same High Temp V2 resin batch, the same 50 μm layer thickness, the same two-stage post-curing protocol, and the same 5 μm vacuum aluminium metallisation provided by AMB Industri AB in a single coating run. Any differences in measured performance between the seven prototypes can therefore be attributed neither to differences in the AM technology, in the polymer material, nor in the metallisation process, but only to variation occurring within these fixed process settings. For example, the amount of unwashed resin during the IPA wash or the variation in the assembly precision of each split block pair.

The observations presented in this thesis can be set in context by comparison with the recent work of Quint et al. (2025), who systematically investigated three metallization processes, physical vapour deposition, electroplating and electroless plating, applied to 3D-printed split-block and monolithic waveguides in the E- and D-bands with a length of 50 mm. Their study found that the choice of metallisation process strongly affects the achievable attenuation, and that for split-block constructions, the lowest losses were obtained when electroplating was used as the final plating step. Two of the investigated samples were printed using a FormLabs Form 3 SLA printer with High Temp resin and metallised using PVD, the process used in the present work, with subsequent copper electroplating. These particular samples showed attenuation comparable to a commercial full-metal waveguide, suggesting that a subsequent layer of copper using electroplating might generate better results.

Considered against the research questions, the results provide the following answers. RQ1: The return-loss criterion of -15 dB at 77 GHz is comfortably met by all seven

prototypes (measured return loss between -24 and -32 dB), but the insertion-loss criterion of 1 dB is not met by any prototype. The lowest measured insertion loss was -3.1 dB for the 50 mm straight section, and the more complex geometries showed substantially higher losses. The process therefore exceeds the target insertion loss by a factor of three or more. RQ2: The parametric simulation showed that the bend radius R has a clear effect on the matching, with R = 0.8 mm yielding a simulated return loss of approximately -30 dB at 77 GHz; the S-bend separation distance D had only a modest effect within the investigated range. The measured return-loss data are consistent with the simulated data, confirming that the bend geometry influences the impedance matching as predicted. RQ3: The fabrication process exerts a strong influence on the measured performance. The measured surface roughness alone, although non-negligible at Rq values comparable to the skin depth, cannot explain the wide variation in insertion loss between prototypes. The observation of residual uncured resin in the longer and more complex channels, together with comparison to Quint et al. (2025), which identifies electroplating rather than PVD as the preferred metallisation for split-block constructions, indicates that process-related mechanisms, such as cleaning quality, split-block seal integrity, and metallisation choice, dominate the observed performance variation. Returning to the project objectives, the cost target of below USD 1000 per prototype is comfortably met, the material and machine cost of an SLA print on the Form 4B with High Temp V2 resin is of the order of a few tens of dollars per part, and the vacuum metallization service from AMB Industri AB adds a similarly modest amount. Considered as a foundational study, the work delivers the design and characterisation data on which SafeRadar can build upon, when extending the process toward full antenna prototypes containing multiple bends, transitions, and radiating elements. The findings should be interpreted as applying to this specific combination of printer, resin, metallisation process and assembly method; generalisation to other AM processes or to other metallisation vendors requires further work.

6.2 Discussion of the Method

The methodological choices made in this work were appropriate for the research questions but introduced several limitations that should be considered when interpreting the results. The electromagnetic simulations in Ansys HFSS 2024 were configured as an idealised best-case scenario, with perfect electric conductor (PEC) walls and lossless vacuum filling. This was a deliberate simplification intended to isolate geometric effects from material- and process-dependent losses. The consequence is that the simulated S-parameters represent an upper bound on the achievable performance and cannot be directly compared with the measured values in absolute terms. The close agreement obtained for the return loss confirms that the geometric features are reproduced accurately by the process, while the lack of agreement obtained for the insertion loss is attributable to physical effects not captured by the idealised model. A more elaborate simulation incorporating finite aluminium conductivity, surface roughness, and a finite-width gap at the split plane could have provided closer agreement with the

measurements. However, such a model requires previous knowledge of these properties, neither of which was known at the beginning of the project.

The surface characterisation followed the workflow specified for compliance with ISO 21920, using stitched measurements of at least 5 mm x 1 mm and three profile extractions per region. The use of three profiles per prototype provides a measure of local variability but is a statistically modest sample; a larger number of profiles or, alternatively, areal parameters from ISO 25178 would have given a more robust characterisation.

Only one prototype was fabricated for each design configuration, and each prototype was measured once. This sampling strategy prioritised investigating the effect of different bend strategies, where no prior data exist in the literature, over statistical confidence in the reproducibility of any single configuration. The chosen strategy was appropriate given the thesis's aim of characterising the dependence on bend geometry as well as the time frame. However, two or three replicates of each configuration would have provided more information on the reproducibility and are recommended for future work.

Due to transport delays of the resin material and isopropyl alcohol, the prototypes were printed with an insufficient amount of isopropyl alcohol, leading to contamination of the solution and resulting in resin residual in the parts discussed in Section 6.1. The washing step proved, therefore, critical for the achievement of residual-free printed prototypes. Given the time frame of the thesis and the transport delays, it would not have been possible to wait for the arrival of the isopropyl alcohol and still manage to perform the metallisation and measurement steps.

The RF measurements were performed with a vector network analyser over the 75-110 GHz range. Although VNA measurement is the established method for two-port characterisation, several measurement-related factors may have influenced the results. The flange interface with the frequency extender was made using the standard UG-387/U alignment, and the measurements were not repeated with multiple disconnect cycles per prototype. Misalignment in the flange interface could produce return loss degradation, and repeated measurements would have allowed the contribution of flange-related variability to be isolated from the inherent performance contribution of the device. However, due to the prototypes being quite fragile and the risk of breakage, as well as the low return loss obtained from all seven prototypes, repeated measurement cycles were therefore ruled out.

6.3 Future Work and Recommendations

The results of this thesis point to several directions in which further work would strengthen the conclusions and increase the practical value of the process for SafeRadar. The most immediate need is to quantify the manufacturing reproducibility of the process. Fabricating several copies of the same design and measuring them multiple times, including repeated disconnect cycles at the flange, would separate the

contributions of print, metallisation, assembly and measurement variations, and would establish the achievable consistency between identical prototypes.

The increase in insertion loss with geometric length and complexity observed in the present results points to mechanisms that accumulate along the prototypes. The split-plane seal is a likely candidate and requires further investigation. Applying a controlled clamping force or introducing a thin subsequent conductive layer at the split-plane could each reduce leakage and improve transmission performance. A systematic study comparing different split-plane designs for additively manufactured waveguides would complement the data presented here.

The metallisation process itself was not varied in this work. As discussed in Section 6.1, the most directly actionable recommendation arising from the present results is therefore to repeat the experiments with an alternative or hybrid metallisation, for example, a thin PVD layer followed by electroplated copper, and to compare the attenuation directly with the PVD-only baseline established here. Future studies could also investigate the effect of coating thickness within and beyond the 3-5 μm range used by AMB, and the effect of alternative coating materials such as silver. Finally, future measurements would benefit from the use of frequency extenders matched directly to the WR-12 standard, allowing measurements across the full WR-12 operating range of 60-90 GHz.

From a long-term perspective, the bend geometry data generated in this thesis are intended to function as the foundational dataset for SafeRadar's future development of in-house antenna prototypes. The characterised waveguide elements can serve as a starting point for further building blocks that can be combined into more complex designs and ultimately into complete antennas. Extending the investigation to other components, such as twists, transitions and antenna apertures, would broaden the practical applicability of the process for in-house prototyping, which is the long-term objective that motivated this initial study.

6.4 Social, Economic, and Environmental Aspects

The motivation for the work is primarily economic, and the long-term goal is to enable in-house development of complete antenna prototypes for 77 GHz automotive radar applications. By enabling prototyping of the cost and lead time of external manufacturing, the process supports faster development at SafeRadar and lowers the threshold for exploring novel antenna designs. The waveguide structures studied in this thesis are the foundational building blocks on which more complex antenna geometries can be developed in subsequent iterations. For a small or medium-sized company, where the cost of each prototyping cycle is a significant fraction of the development budget, an internal capability of this kind shortens the time from concept to validated design and therefore contributes to competitiveness in the automotive radar sector.

From an environmental perspective, the vacuum metallisation process used in this work is described by the supplier as solvent-free and free of harmful baths, in

contrast to electrolytic processes such as electroplating (AMB Industri AB, n.d). The SLA printing process itself produces relatively little waste compared with subtractive machining, which discards a large fraction of the starting material as chips. The post-curing and metallisation steps consume energy, but the small size and short cycle time of the prototype parts limit the absolute consumption per part. A full life-cycle assessment comparing in-house additive manufacturing with external CNC machining is beyond the scope of this thesis but would be a relevant complement for a future study.

A target conflict that arises in the present work is the trade-off between cost and RF performance. The chosen process meets the cost target with substantial margin and delivers RF performance acceptable for compact components, but the insertion loss grows faster than geometric scaling predicts for the longer and more complex structures. For prototyping and early-stage iterative design, where the goal is to explore many variants quickly and where the absolute loss budget is less critical, this trade-off is considered acceptable. For final production components delivered to end customers, however, the higher loss would generally be unacceptable, and the conventional manufacturing route would remain the preferred choice. The process investigated here should be regarded as a complement to, rather than a replacement for, the established manufacturing chain.

7. Conclusions

The aim of this thesis was to develop and evaluate a cost-effective manufacturing method for 77 GHz waveguide structures based on additive manufacturing combined with metallisation, as a foundation for SafeRadar's future in-house antenna prototype development. The work delivered a documented process and characterised seven 77 GHz prototypes across straight, single 90-degree bend, and S-bend configurations.

The investigated process produces WR-12 prototypes with excellent impedance matching, satisfying the -15 dB return-loss criterion across all configurations (measured return loss between -24 and -32 dB at 77 GHz), but does not meet the 1 dB insertion-loss criterion. The lowest measured insertion loss was 3.1 dB, rising to 17.4 dB for the more complex geometries. The bend geometry influenced the matching as predicted, with R = 0.8 mm identified as the optimal radius. The manufacturing process emerged as the dominant influence on insertion loss: the measured surface roughness was comparable to the skin depth of aluminium at 77 GHz, while residual resin in the longer and more complex channels coincided exactly with the prototypes that exhibited the highest losses.

The cost target of below USD 1000 per prototype was comfortably met. Considered as a foundational study, the work delivers parametric design data and a characterised process baseline that SafeRadar can build upon when extending the route toward complete antenna prototypes. The principal lessons are that polymer-based additive manufacturing is viable for compact 77 GHz waveguide components, but that low insertion loss will require improved cleaning protocols and an alternative or subsequent metallisation strategy such as electroplating.

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Appendices

Appendix 1

Table 1. *Rq and Ra values of the metallised waveguide channels, calculated from all three profiles per prototype.*

	Extracted Profile nr 1		Extracted Profile nr 2		Extracted Profile nr 3	
Prototype	Rq (µm)	Ra (µm)	Rq (µm)	Ra (µm)	Rq (µm)	Ra (µm)
Straight - 50 mm	0.29	0.23	0.28	0.22	0.28	0.20
Straight - 100 mm	0.28	0.22	0.25	0.18	0.15	0.11
Single bend - R = 0.8	0.29	0.22	0.25	0.16	0.23	0.16
Single bend - R = 2	0.23	0.15	0.20	0.15	0.22	0.14
Single bend - R = 3	0.24	0.17	0.20	0.15	0.23	0.16
S-shaped - D = 2	0.29	0.24	0.24	0.18	0.25	0.17
S-shaped - D = 3	0.26	0.18	0.18	0.14	0.25	0.18

Appendix 2

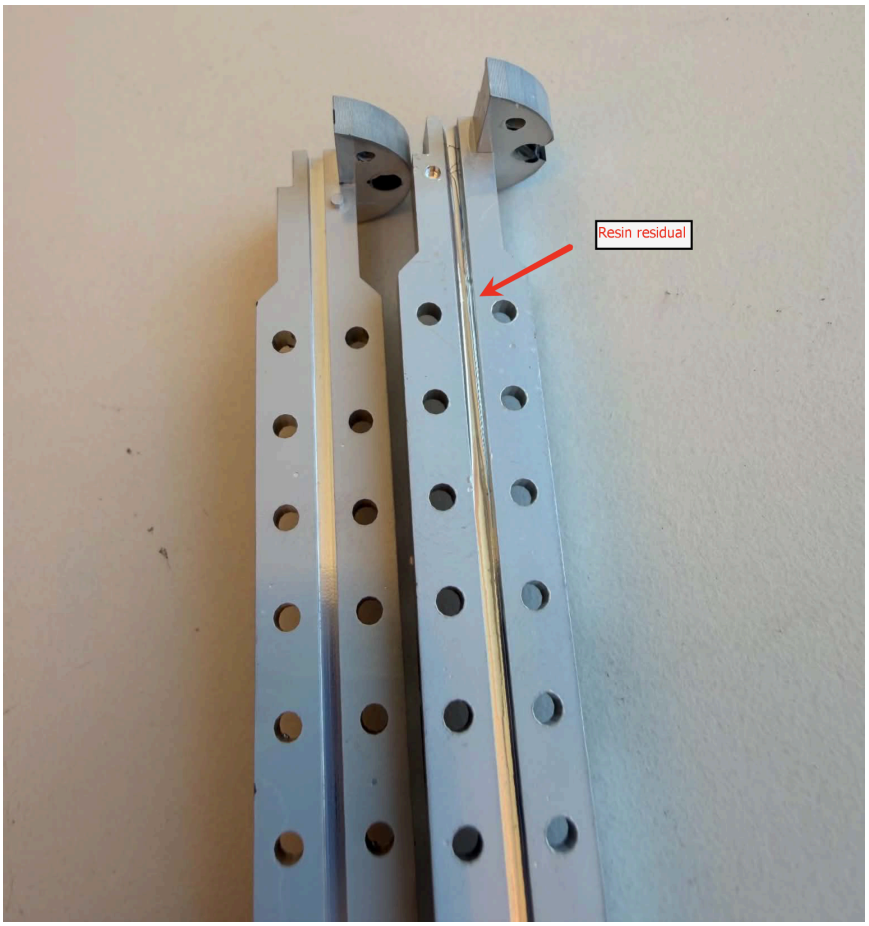


Figure 1: *Straight 100 mm waveguide prototype, showing resin residual in the channel of one of the halves.*

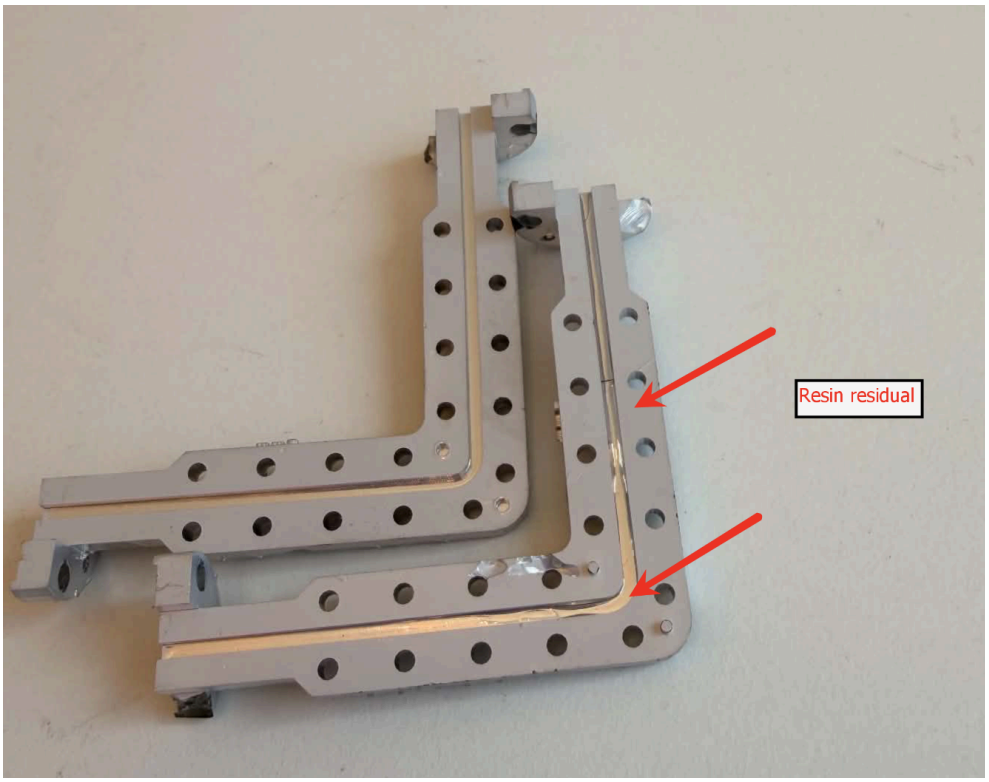


Figure 2: *Single bend - R = 2mm waveguide prototype, showing resin residual in the channel of one of the halves.*

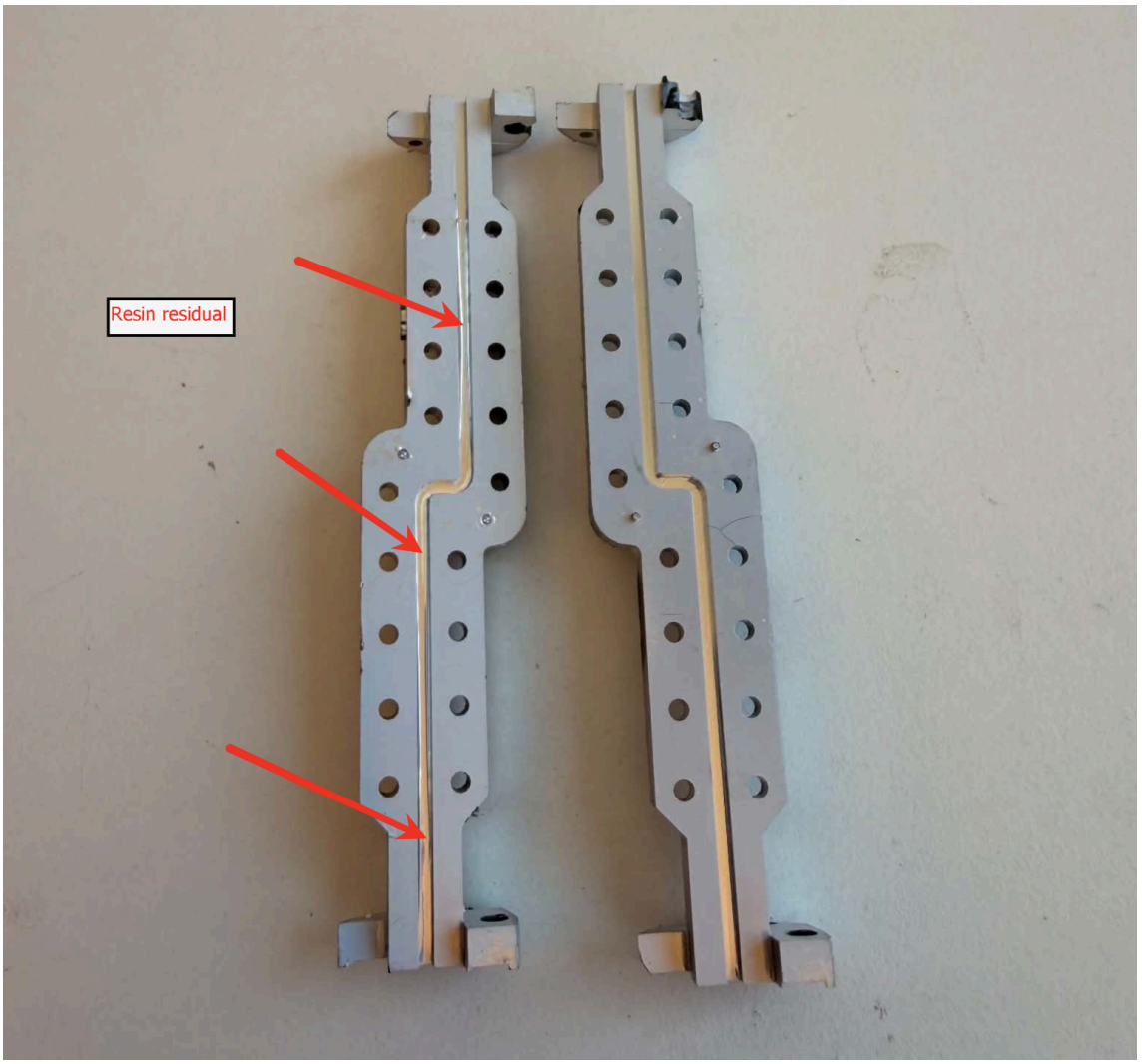


Figure 3: *S-shaped - D = 3mm waveguide prototype, showing resin residual in the channel of one of the halves.*